

FPGA Development for Radar, Radio-Astronomy and Communications

THE
RADAR
MASTERS COURSE



Dept. Electrical Engineering, University of Cape Town
Private Bag, Rondebosch, 7701, South Africa
<http://www.rmsg.uct.ac.za>



Presented by Dr John-Philip Taylor
Convened by Dr Stephen Paine

Day 3 – 5 August 2026

Outline

Memory-Mapped Bus

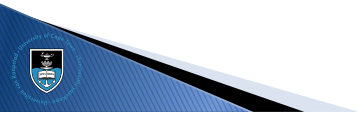
Advanced Clocking

Clock Domains

Pipelines

Streaming Processors

Flow Control



Outline

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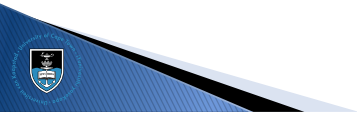
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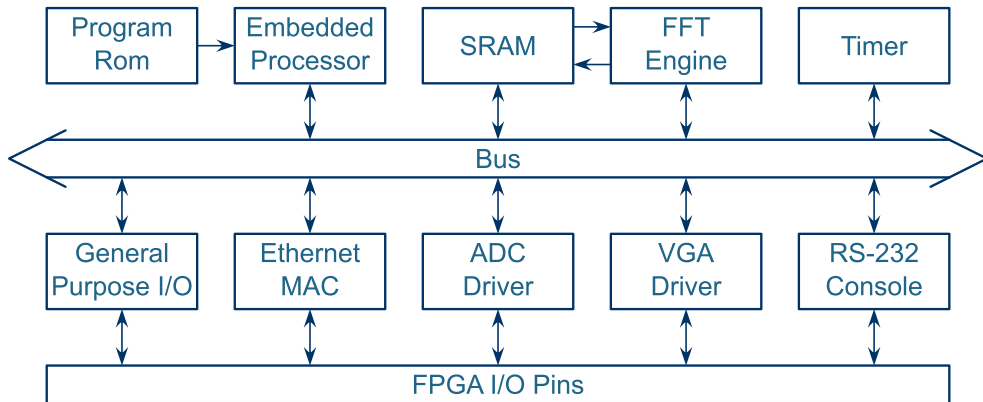
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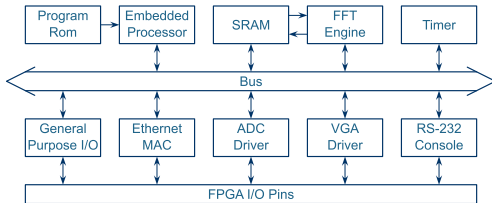
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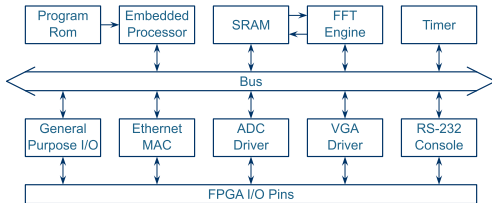
Memory-Mapped Bus



- ▶ Every node on the bus has an address range allocated
- ▶ Generally used for writing control registers, reading system status and accessing memory
- ▶ Altera Qsys uses Avalon
- ▶ Xilinx IP Integrator and ARM processors use AXI
- ▶ Many open-source projects use Wishbone



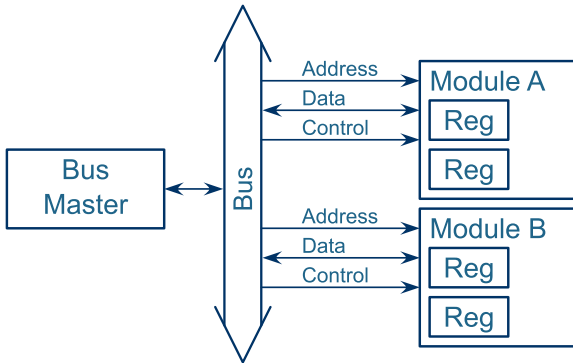
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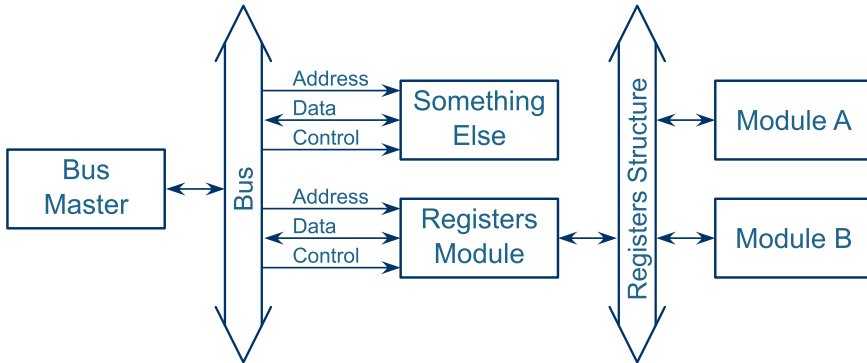
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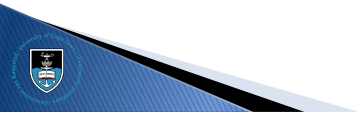
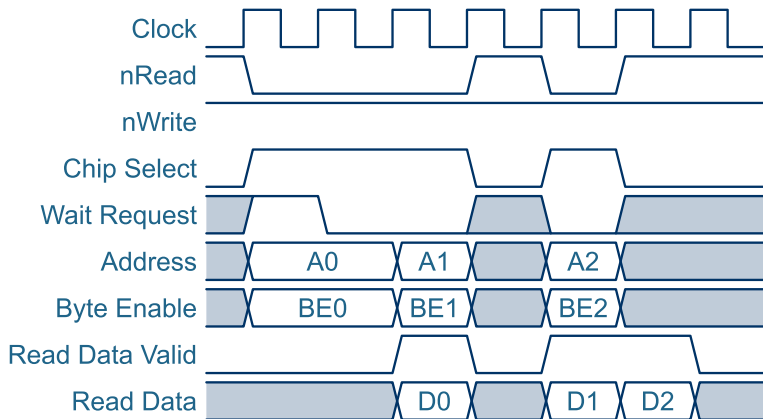
Memory-mapped Architectures



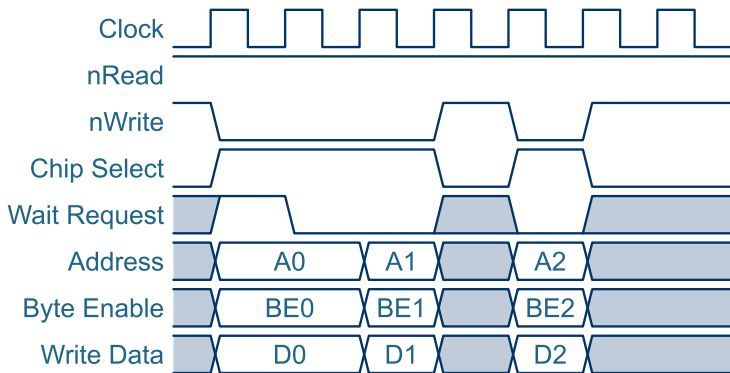
Memory-mapped Architectures



Interface Timing Diagrams



Interface Timing Diagrams

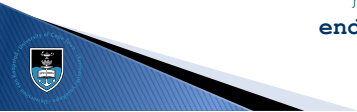


Structures

- ▶ SystemVerilog only
- ▶ Define these once in a separate file, typically as part of a package

```
package Structures;
  typedef struct{
    logic [ 1:0]Buttons;
    logic [ 9:0]Switches;
    logic [31:0]StatusBits;
  } RD_REGISTERS;

  typedef struct{
    logic [ 9:0]LEDs;
    logic [31:0]NCO_Frequency;
    logic [ 2:0]Bandwidth;
  } WR_REGISTERS;
endpackage
```



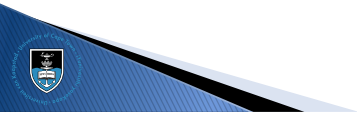
Structures

```
import Structures::*;

module Registers(
    input ipClk, ipReset,

    input  [15:0] ipAddress,
    output [31:0] opRdData,
    input  [31:0] ipWrData,
    input                ipWrEn,

    input  RD_REGISTERS ipRdRegisters,
    output WR_REGISTERS opWrRegisters
);
...
```



Structures

```
// In the top-level module...

RD_REGISTERS RdRegisters;
WR_REGISTERS WrRegisters;

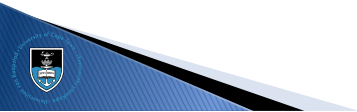
Registers Registers_Inst(
    Clk, Reset,
    Address, RdData, WrData, WrEn,
    RdRegisters, WrRegisters
);

NCO NCO_Inst(
    Clk, Reset,
    WrRegisters.NCO_Frequency,
    RdRegisters.StatusBits[15]
);
```



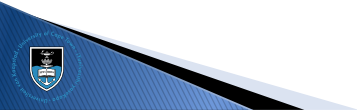
Structures

- ▶ You can map a structure to an input port and then use only what you need within the submodule
- ▶ You can not map the same structure to an output port of more than one module, i.e. you need to map the structure members individually
- ▶ Luckily, most registers are control registers, and therefore input ports of the target modules



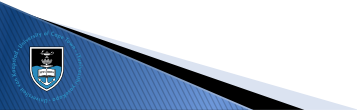
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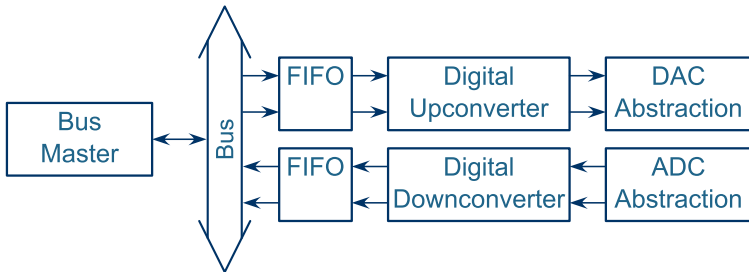


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Memory-mapped Streams



- ▶ It is often convenient to allocate a bus address to a stream
- ▶ A write to that address injects one unit into the stream
- ▶ A read from that address reads one unit from the stream
- ▶ Most often unidirectional and FIFO-buffered with feed-back registers



Platform Designer

Practical 06 – Platform Designer introduces the Platform Designer

The screenshot displays the Platform Designer (QSys) interface for a project named 'C:\PT\06\HUB\UCT-FPGA-Course-2024\Tutorial\Qsys\Qsys1'. The interface is divided into several panels:

- IP Catalog:** Shows a tree view of available IP components. The 'Avalon-MM Pipeline Bridge' is highlighted under the 'Memory Mapped' category.
- System Contents:** Displays a hierarchical view of the system components. The 'Avalon-MM Pipeline Bridge' is selected, and its connections are shown in a diagram.
- Parameters:** Shows the configuration parameters for the 'Avalon-MM Pipeline Bridge'. The parameters are grouped into sections:
 - General:** Data width (32), Symbol width (8), Address width (30), Address unit (WORDS), Burst (Maximum burst size (words): 1), Line wrap bursts (checked).
 - Read/Write:** Maximum pending read transactions (4), Pipeline command signals (checked), Pipeline response signals (checked).
 - Response:** Use Avalon Transaction Responses (unchecked).

The bottom status bar shows '0 Errors, 0 Warnings' and buttons for 'Generate HDL...' and 'Finish'.



Platform Designer

Practical 06 – Platform Designer introduces the Platform Designer

System: QSys Path: master						
Use	Connections	Name	Description	Export	Clock	Base
☒		clk_0	Clock Source			
		clk_in	Clock Input	clk	exported	
		clk_in_reset	Reset Input	reset		
		clk	Clock Output	Double-click to export	clk_0	
		clk_reset	Reset Output	Double-click to export		
☒		JTAG	JTAG to Avalon Master Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		clk_reset	Reset Input	Double-click to export		
		master	Avalon Memory Mapped Master	Double-click to export	[clk]	
		master_reset	Reset Output	Double-click to export		
☒		master	Avalon-MM Pipeline Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		reset	Reset Input	Double-click to export	[clk]	
		s0	Avalon Memory Mapped Slave	master	[clk]	
		m0	Avalon Memory Mapped Master	Double-click to export	[clk]	
☒		SDRAM	Avalon-MM Pipeline Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		reset	Reset Input	Double-click to export	[clk]	
		s0	Avalon Memory Mapped Slave	Double-click to export	[clk]	0x0000_0000
		m0	Avalon Memory Mapped Master	sdram	[clk]	
☒		Registers	Avalon-MM Pipeline Bridge			
		clk	Clock Input	Double-click to export	clk_0	
		reset	Reset Input	Double-click to export	[clk]	
		s0	Avalon Memory Mapped Slave	Double-click to export	[clk]	0x0400_0000
		m0	Avalon Memory Mapped Master	registers	[clk]	

Outline

Memory-Mapped Bus

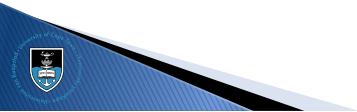
Advanced Clocking

Clock Domains

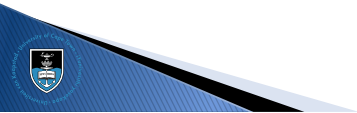
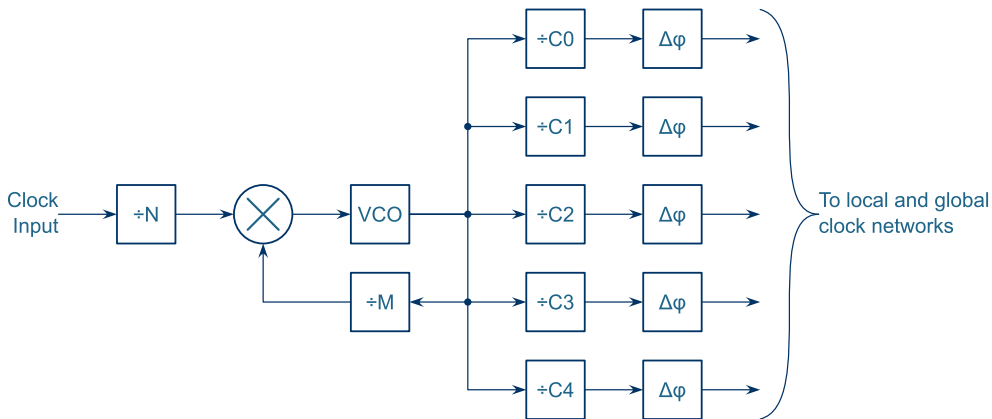
Pipelines

Streaming Processors

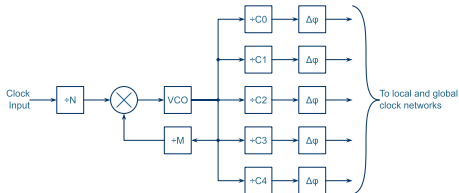
Flow Control



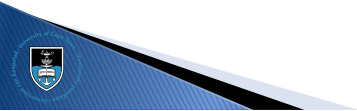
Phase-locked Loop



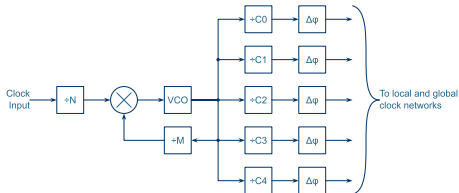
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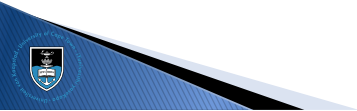
- ▶ Useful for generating a range of related clocks
- ▶ Output clock rising and falling edges can be set, independently, to a resolution equal to the VCO period, which is typically about $1 \text{ ns} \pm 500 \text{ ps}$
- ▶ All output clocks can be considered to be in the same clock domain
- ▶ Noise sensitive applications should avoid PLLs due to high phase noise



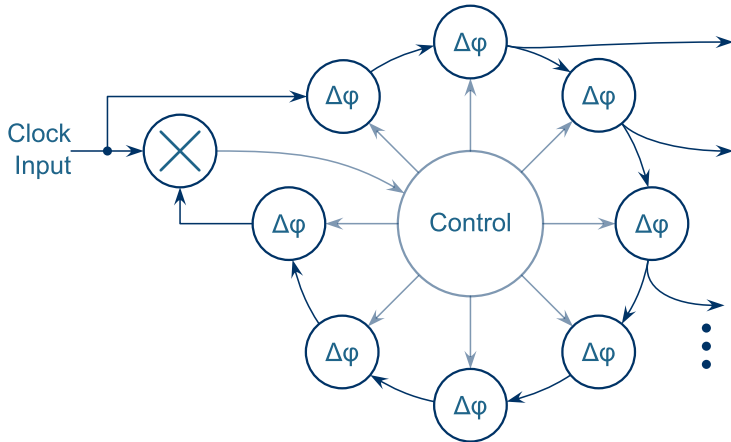
PLL Compensation



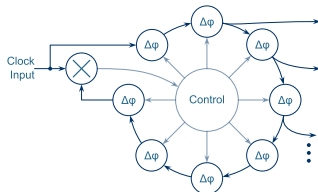
- ▶ **Direct:** No compensation, which results in better jitter performance
- ▶ **Normal:** The clock at the register is phase-aligned with the input clock pin
- ▶ **Source Synchronous:** The PLL ensures that the data delay from pin to register and the apparent clock delay from pin to register is the same
- ▶ **Zero-Delay Buffer:** The output clock pin is phase-aligned with the input clock pin



Delay-locked Loop

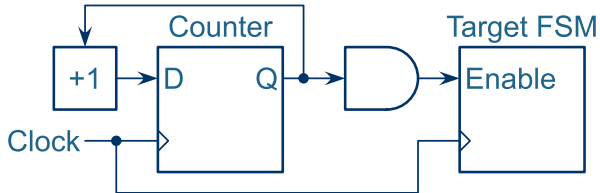


Delay-locked Loop

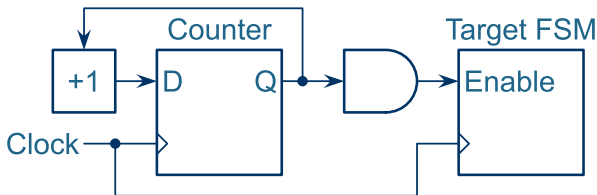


- Useful for performing phase compensation when interfacing to fast peripherals
- All output clocks can be considered to be in the same clock domain as the input clock

Clock-Enable Generated

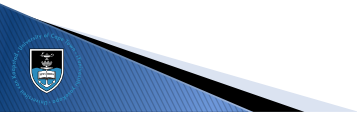
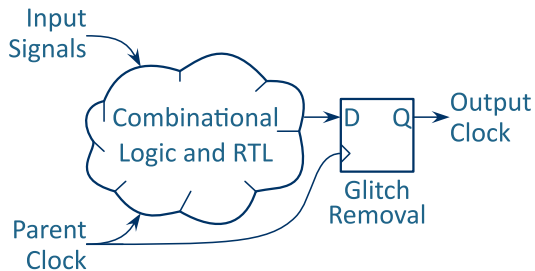


Clock-Enable Generated

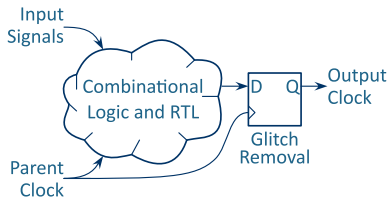


- ▶ Useful when mixing fast and slow systems
- ▶ Potential to reduce overall device power usage
- ▶ Cannot be used for clocking external devices
- ▶ The multi-cycle timing requirements must be manually specified

Ripple and Gated Clocks

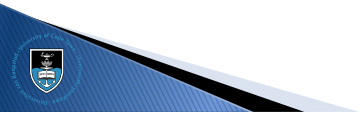
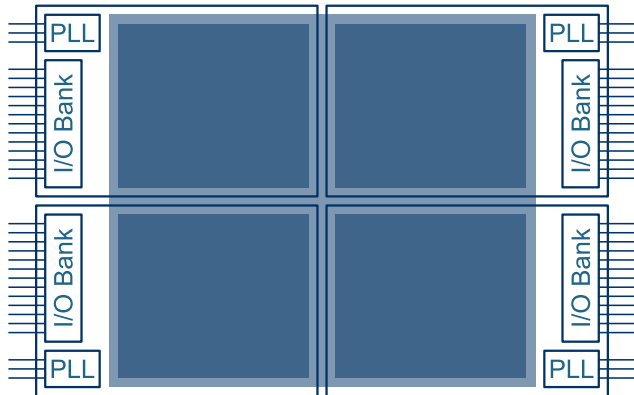


Ripple and Gated Clocks

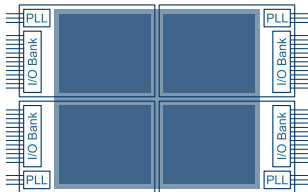


- ▶ Unrelated to all other clocks in the system, including the parent clock
- ▶ Often necessitates complex clock-domain crossing schemes to the general system clock
- ▶ Useful when:
 - ▶ Requiring very low power usage
 - ▶ The frequency must change often
 - ▶ The clock must drive an external device
 - ▶ etc.

Clock Regions



Clock Regions



- ▶ FPGAs are generally organised into regions, each with its own PLL and I/O bank
- ▶ The local clock network is more efficient (and faster) than the global network
- ▶ When laying out the PCB, keep fast I/O in the same region as their clock

Outline

Memory-Mapped Bus

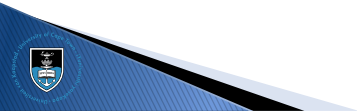
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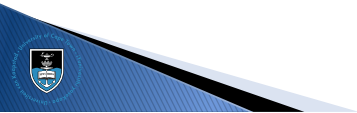
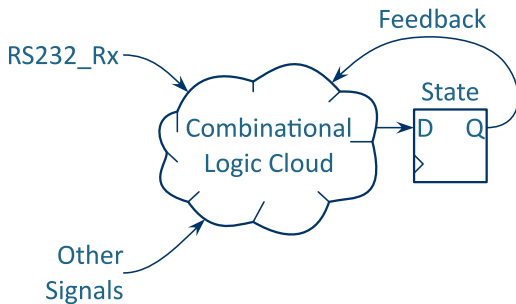


Common Mistake

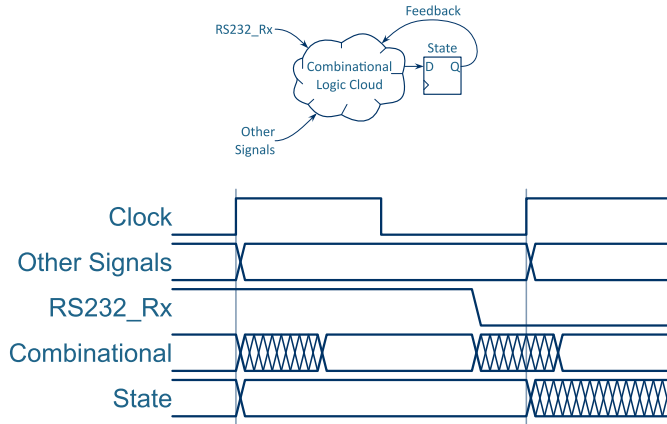
```
module RS232(  
    input  ipClk, ipReset,  
    input  ipRS232_Rx, // Asynchronous external signal  
    output [7:0] opData  
);  
  
... // Definitions, state machine boilerplate, etc.  
  
Idle: begin  
    if(!ipRS232_Rx) begin  
        State <= Receiving;  
    end  
end  
  
Receiving: begin  
    ...  
end
```



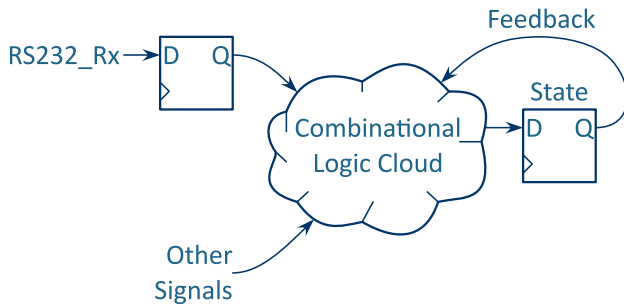
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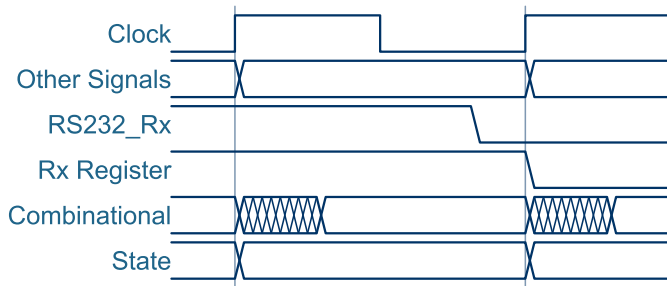
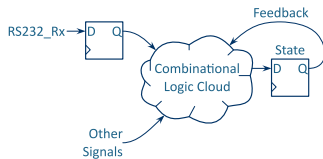
Common Mistake



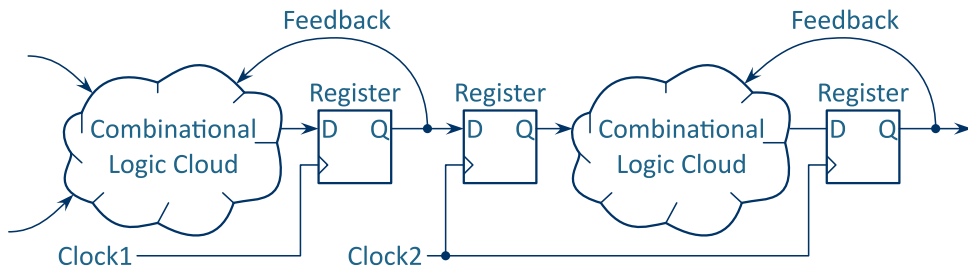
The Solution



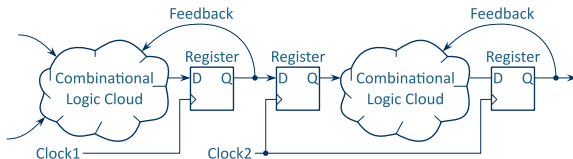
The Solution



Register Chain

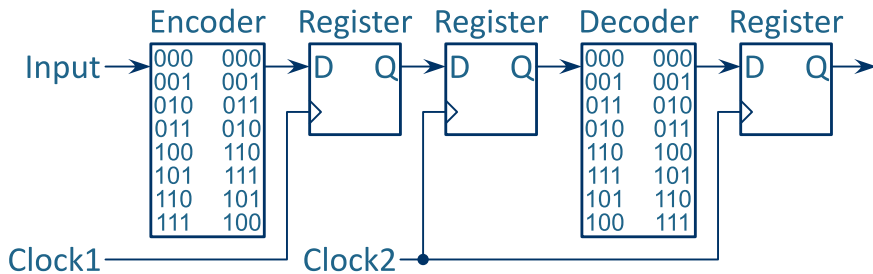


Register Chain

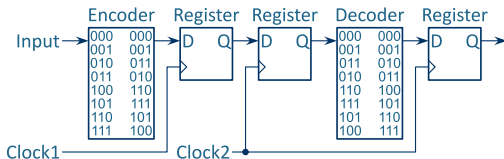


- ▶ Essential for crossing asynchronous external signals
- ▶ Only useful for crossing one bit at a time (unless the system is tolerant to temporary errors)
- ▶ Often used to cross hand-shaking signals
- ▶ Works in both directions (fast to slow / slow to fast)

Gray Coding

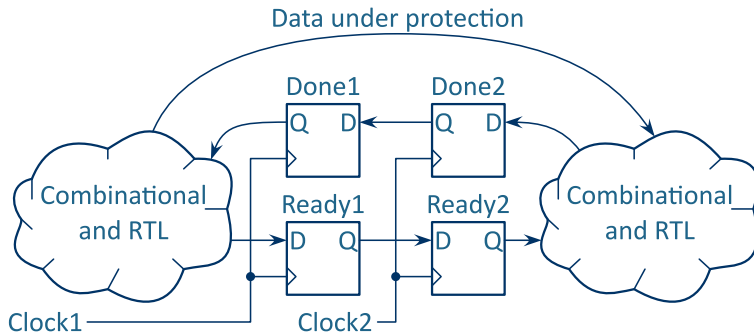


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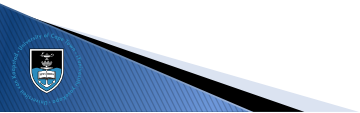
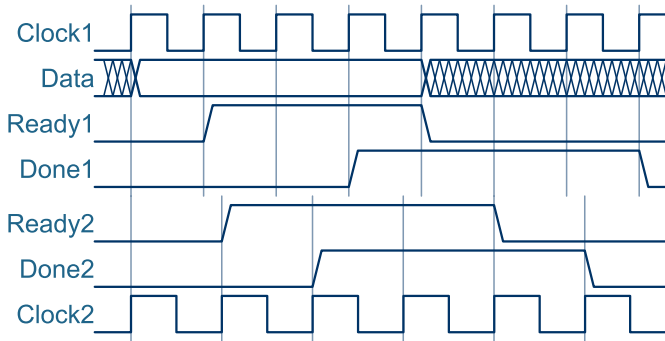


- ▶ Encoded such that only one bit changes at a time
- ▶ Only works for counter data
(FIFO queue pointers, rotary encoders, etc.)
- ▶ Works in both directions (fast to slow / slow to fast)

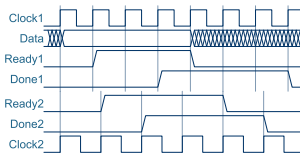
Hand-Shaking



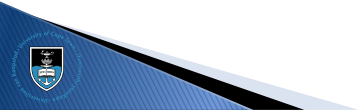
Hand-Shaking



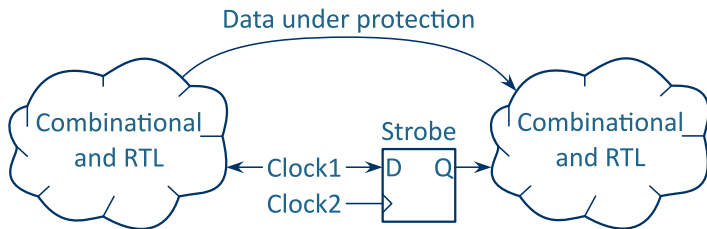
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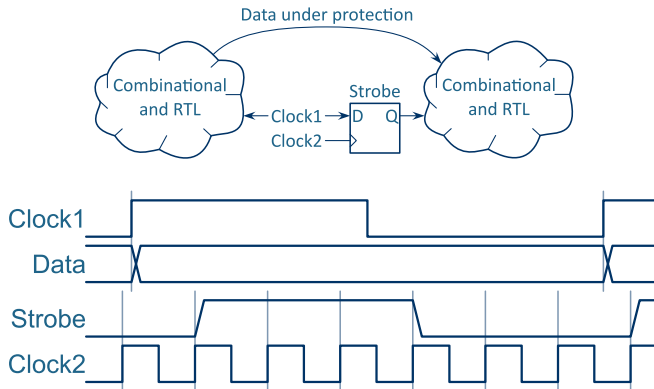
- ▶ Source must guarantee data stability while the handshaking is taking place
- ▶ Not time-efficient: need to wait for the entire transaction sequence to finish before starting the next one
- ▶ Use only for data that does not need to be crossed often
- ▶ Works in both directions (fast to slow / slow to fast)



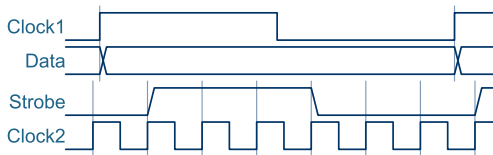
Strobe-based



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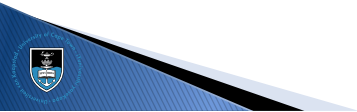


- ▶ The data is valid for as long as the strobe is high
- ▶ As fast as the source clock
- ▶ The strobe edges can be used to perform flow-control
- ▶ Only works when the destination clock is much faster than the source clock
- ▶ Note that the source “clock” can be a strobe signal generated by the source state machine...

FIFO Queues



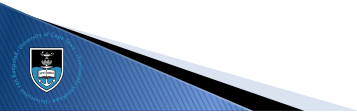
- ▶ Flow-control signals are crossed separately
- ▶ Control signal crossing latency can be hidden in the length of the queue
- ▶ Mixed-width RAM can be used to keep the data-rate constant across different frequencies (especially useful in DDR-based external memory)



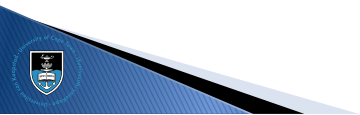
Clock Groups

- Unrelated clock domains must be kept separate in the timing constraints as well

```
set_clock_groups -logically_exclusive \  
-group [get_clocks ADC_CLK_10]      \  
-group [get_clocks MAX10_CLK1_50]   \  
-group [get_clocks MAX10_CLK2_50]   \  
-group [get_clocks {opClk_SDRAM *altpll_component*}]
```



Coffee Break...



Outline

Memory-Mapped Bus

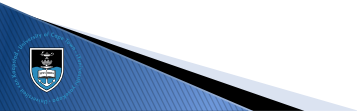
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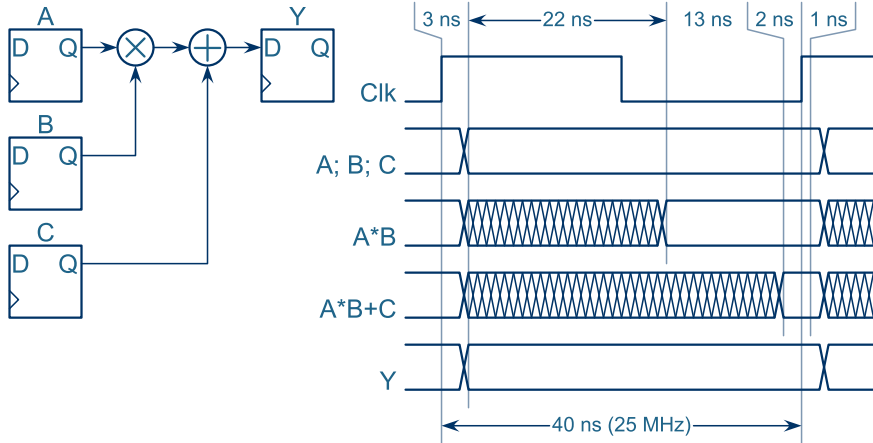
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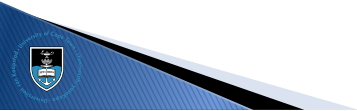
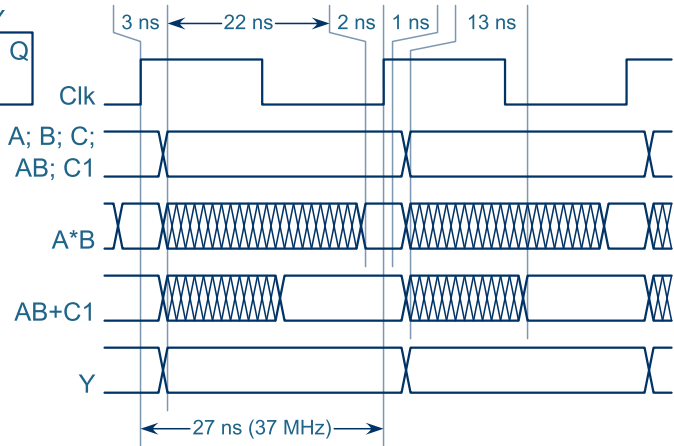
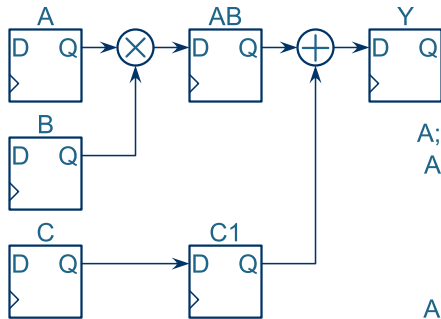
Flow Control



Simple Pipelines



Simple Pipelines



Simple Pipelines

- ▶ Gain throughput at the cost of latency and resources
- ▶ Often easier to use arrays, especially with long chains
- ▶ Keep the index equal to the stage which assigns the value

```
always @(posedge ipClk) begin  
    // Stage 1  
    AB <= A*B;  
    C1 <= C;  
  
    // Stage 2  
    Y <= AB + C1;  
end
```



Simple Pipelines

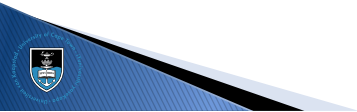
- ▶ Gain throughput at the cost of latency and resources
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```
reg [7:0]C[1:0];

always @(posedge ipClk) begin
    // Stage 0
    C[0] <= C_Input;

    // Stage 1
    AB    <= A*B;
    C[1] <= C[0];

    // Stage 2
    Y <= AB + C[1];
end
```



Simple Pipelines

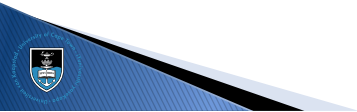
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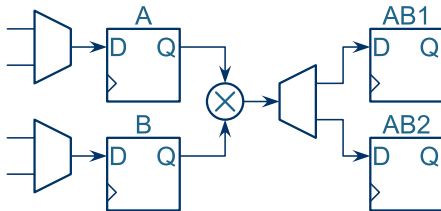


Sharing Resources

```
reg [ 7:0]A, B;
wire [15:0]AB = A * B;
```

```
State1: begin
  AB2  <= AB;
  A    <= A1;
  B    <= B1;
  State <= State2;
end
```

```
State2: begin
  AB1  <= AB;
  A    <= A2;
  B    <= B2;
  State <= State1;
end
```



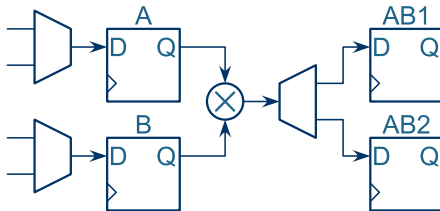
► Gain resource efficiency at the cost of throughput

Sharing Resources

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```
State2: begin
  AB1  <= AB;
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```



- Gain resource efficiency at the cost of throughput

Outline

Memory-Mapped Bus

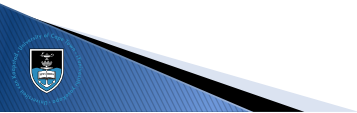
Advanced Clocking

Clock Domains

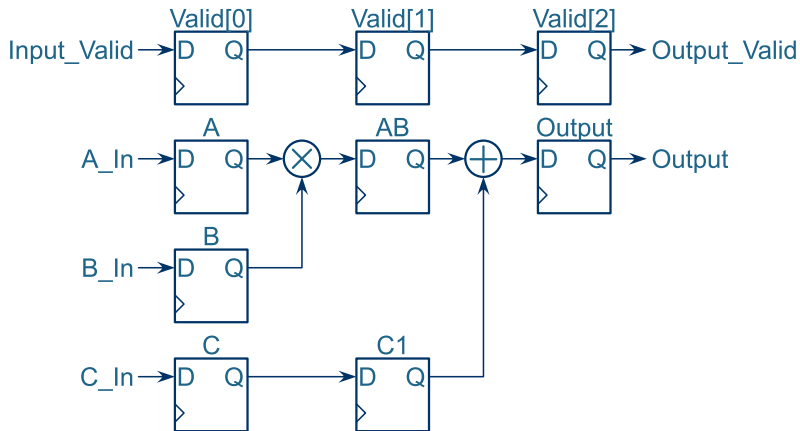
Pipelines

Streaming Processors

Flow Control



Stream Pipeline



Stream Pipeline

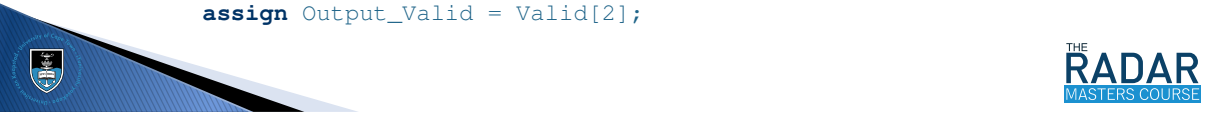
```
reg [2:0]Valid;

always @(posedge ipClk) begin
    // Input
    A      <= A_In; B <= B_In; C <= C_In;
    Valid <= {Valid[1:0], Input_Valid};

    // Stage 1
    AB <= A*B;
    C1 <= C;

    // Stage 2
    Output <= AB + C1;
end

assign Output_Valid = Valid[2];
```



Back-pressure

```
reg [2:0]Valid;

always @(posedge ipClk) begin
    if(Output_Ready) begin
        // Input
        ...

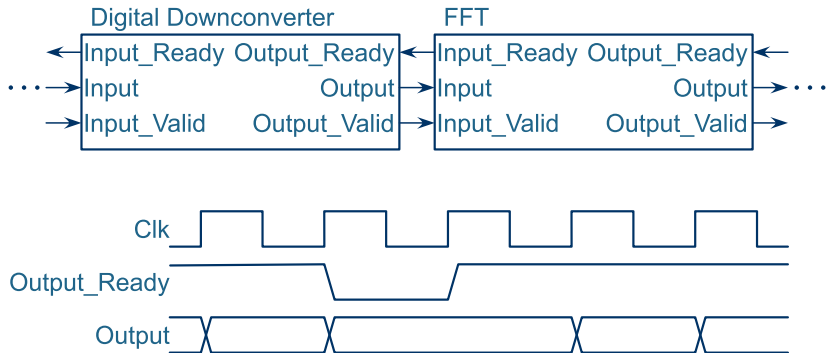
        // Stage 1
        ...

        // Stage 2
        Output <= AB + C1;
    end
end

assign Input_Ready  = Output_Ready;
assign Output_Valid = Valid[2];
```



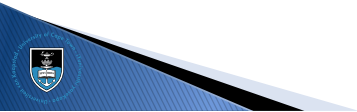
Streaming Processor



Handshaking Strategies

Different buses have different handshaking strategies:

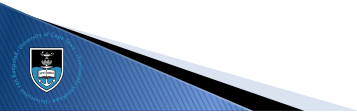
- ▶ An Avalon source is allowed to wait for the sink to be ready before asserting the `Valid`.
- ▶ An AXI sink is allowed to wait for the source to be valid before asserting its `Ready`.
- ▶ A Wishbone slave must wait for valid data from the master before asserting its `Ack`.
- ▶ Do not connect an Avalon source to an AXI sink.



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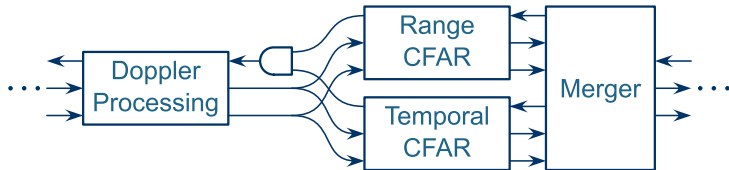


FIFO Queues



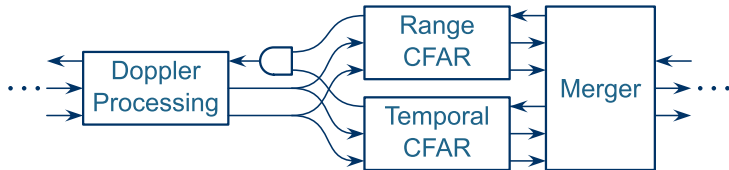
- ▶ Back-pressure makes it possible to move all the FIFO queues into a single queue
- ▶ Generally put the queue where the least amount of data is (saves resources)
- ▶ DSP chains have processing gain, so place the FIFO early in the chain

Complex Structures



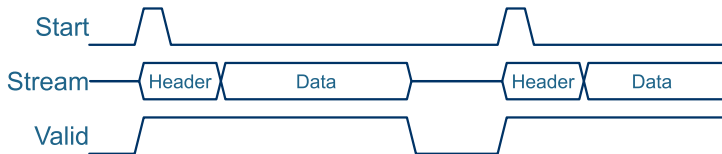
- ▶ Gate the “Valid” with the “Ready”
(if either sink is not ready, both sinks must see a “not valid” input)
- ▶ The “Merger” can take various forms:
 - ▶ Interleave
 - ▶ Alternate (sent one logical unit from the one, then the other, then repeat)
 - ▶ Combine through calculation
 - ▶ etc.

Complex Structures



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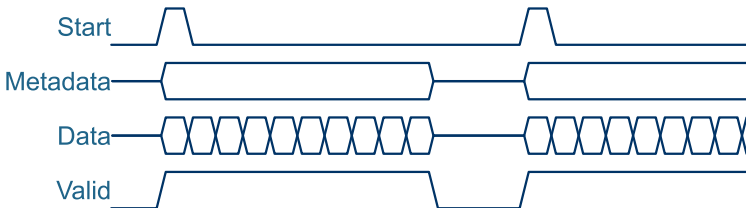
Packet-based Streams



- ▶ Add a start-of-packet strobe (valid for one clock-cycle only)
- ▶ Natural fit for RADAR: one packet per PRI
- ▶ Natural fit for packet-based communication, e.g. Ethernet or UDP (makes it easy to implement Ethernet-based FPGA-in-the-loop testing)
- ▶ The header can contain all sorts of metadata...



Packet-based Streams



- ▶ Metadata can also be implemented as a side-channel
- ▶ The metadata is stable for the duration of the packet
- ▶ No need to extract and / or add the header
- ▶ Easier to filter the stream

Outline

Memory-Mapped Bus

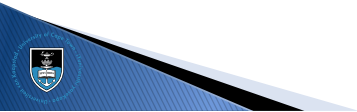
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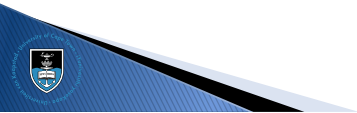
Flow Control



Fire and Forget



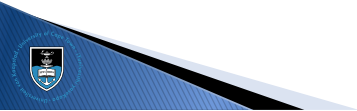
- ▶ No feedback from the destination to the source
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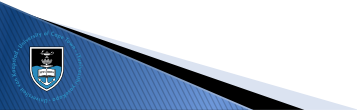
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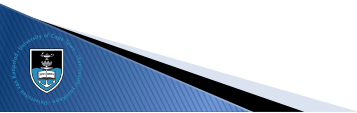
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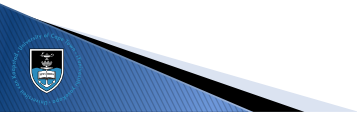
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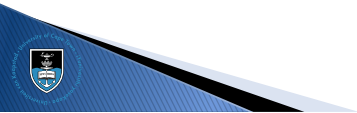
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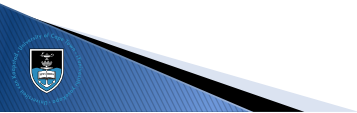
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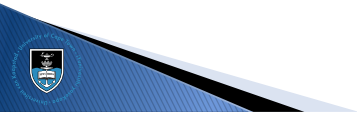
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Command-Response



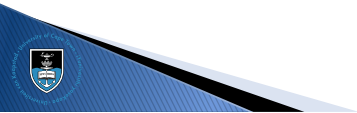
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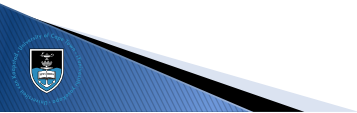
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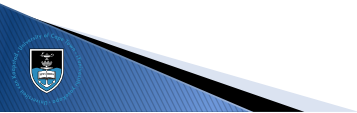
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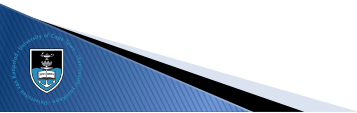
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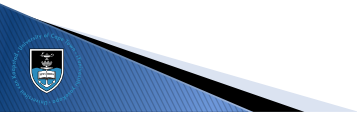
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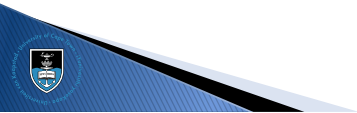
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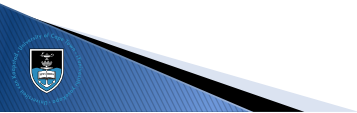
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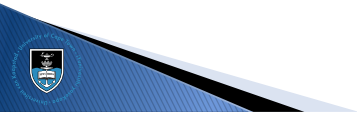
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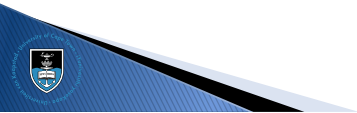
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Sliding Window



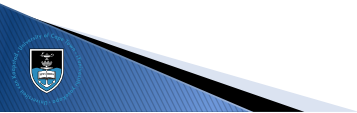
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- ▶ This is used in TCP, among other protocols



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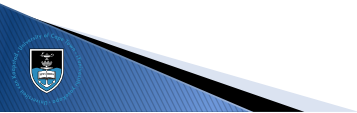
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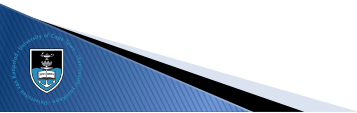
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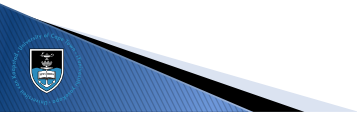
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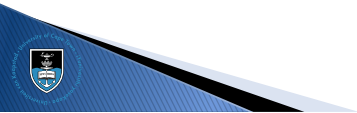
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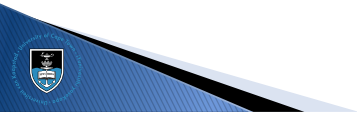


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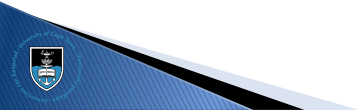
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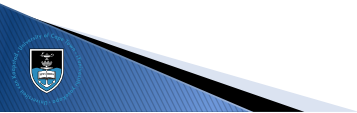
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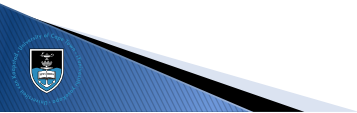
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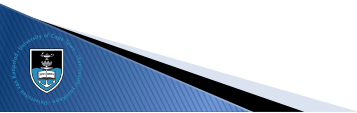
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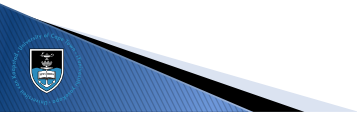
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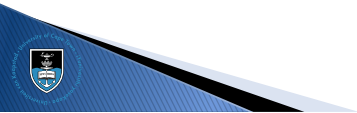
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Credit Based (1)



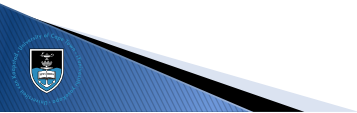
- ▶ The source assumes that the destination has a certain space
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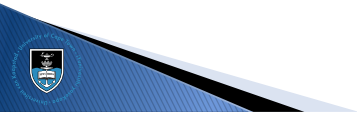
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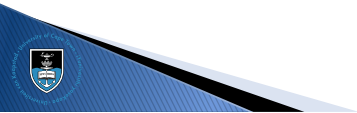
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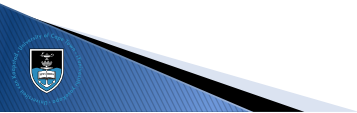
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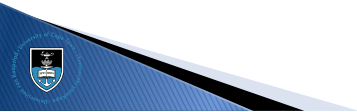
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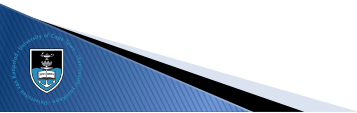
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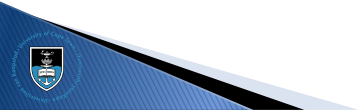
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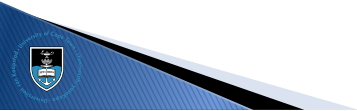
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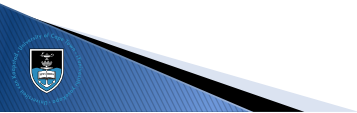
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Credit Based (1)



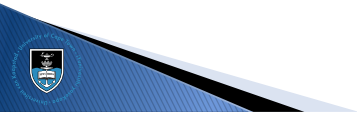
- ▶ The source assumes that the destination has a certain space
- ▶ The destination acknowledges all incoming packets
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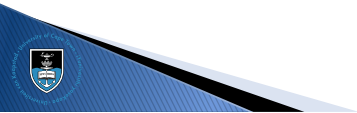
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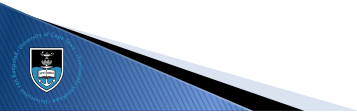
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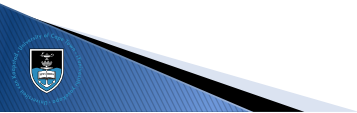
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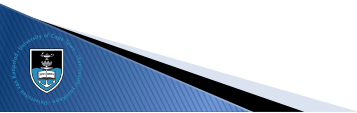
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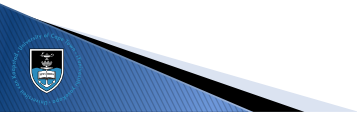
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- ▶ This is used in PCIe, among other protocols



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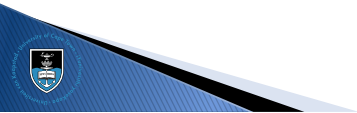
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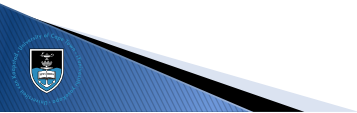
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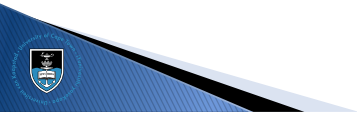
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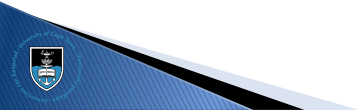
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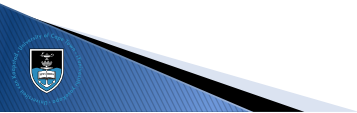
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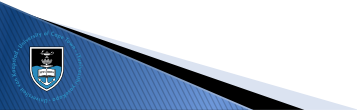
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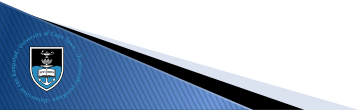
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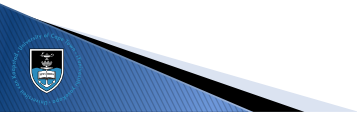
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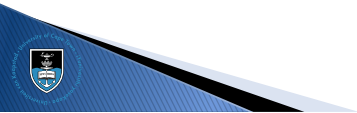
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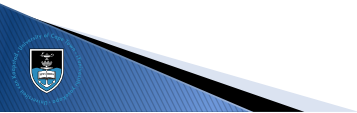
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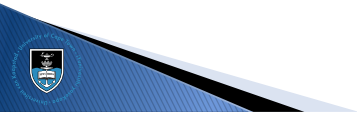
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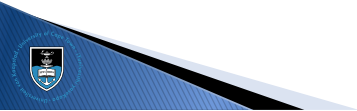


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Presented by Dr John-Philip Taylor
Convened by Dr Stephen Paine

Day 3 – 5 August 2026